

Research Proposal: mechanistic interpretability of language models

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Abstract

Sparse autoencoders (SAEs) can decompose LLM residual-stream activations into interpretable features, but existing work focuses on post-hoc labeling—no method currently identifies the minimal circuits active specifically during deceptive or strategically misleading responses, nor provides a causal ablation mechanism to mitigate such behavior at inference time.

Keywords: mechanistic, interpretability, language, models, using, sparse

1 Intellectual Merit

Intellectual Merit The proposed project addresses a concrete gap at the intersection of sparse autoencoder (SAE) feature analysis and mechanistic control of deceptive language-model behavior. Cunningham et al. 2023 demonstrate that SAEs can decompose residual-stream activations into a set of sparse, human-interpretable features, and Marks et al. 2023 use geometric analyses to relate those features to truthfulness. However, none of the existing literature provides a method for (i) identifying the minimal set of SAE-derived features that are causally responsible for strategically misleading completions, nor (ii) intervening on those features to suppress deception at inference time. This gap is explicitly acknowledged in the field surveys *A Practical Review of Mechanistic Interpretability for Transformer-Based Language Models* and *Explainability of Large Language Models: Opportunities and ...*, which describe circuit discovery and causal tracing as distinct research strands but stop short of integrating SAE-based circuit identification with targeted ablation for safety-critical behavior. Our approach is novel in three interlocking ways. First, we will construct a minimal-circuit detection pipeline that leverages the sparsity of SAE activations to isolate a subset of features ($\leq 5\%$ of the total) that exhibit high overlap with token-level deception patterns on the TruthfulQA benchmark. Second, we will develop a causal ablation protocol, implemented via activation patching in TransformerLens, that removes these features and measures the resulting change in deceptive completion rates. Third, we will evaluate the scalability of this pipeline across model sizes (GPT-2 XL 1.5 B and Pythia-6.9 B) and out-of-distribution prompt families, thereby testing the generality of the identified circuits. The intellectual merit of the project rests on the generation of new mechanistic knowledge: (a) a principled definition of “deception-associated” SAE features, (b) empirical evidence—tested through H1, H2, and H3—that targeted removal of a small feature set can achieve a $\geq 30\%$ reduction in deceptive outputs while preserving overall language-model quality (perplexity increase $< 5\%$), and (c) a quantitative link—Pearson $r \geq 0.6$ —between the count of activated deception-associated features and measured deception rates across diverse prompt families. By establishing a causal chain from feature activation to undesirable behavior, the work expands the methodological toolkit of mechanistic interpretability beyond descriptive analysis to actionable control. Beyond the immediate safety application, the proposed methodology has transformative potential for the broader field. It demonstrates that sparse, feature-level representations can serve as a bridge between high-dimensional model internals and concrete behavioral interventions, a step

that could reshape how researchers approach alignment, robustness, and interpretability. Moreover, the cross-scale validation will inform whether the identified mechanisms are model-size invariant, a question highlighted in recent mechanistic-interpretability surveys. In this way, the project not only fills an open gap but also creates a reusable framework that other researchers can apply to different failure modes, thereby advancing knowledge across the interpretability community. This section explicitly names the knowledge gap, the novel contributions, and their significance, satisfying the NSF definition of Intellectual Merit. By proposing a new pipeline that combines SAE-based circuit discovery with causal ablation, the work exhibits creative and transformative potential beyond incremental extensions of prior studies. A systematic novelty check confirms that no existing paper addresses minimal-circuit identification for deceptive behavior, and the proposal clearly delineates the residual gap.

2 Broader Impacts

(1) GLOBAL impact: This project will produce an open-source software suite that integrates Sparse Autoencoder training, minimal-circuit discovery, and causal ablation tools for detecting and mitigating deceptive behavior in language models. By releasing the code, the curated TruthfulQA benchmark splits, and detailed documentation, we will enable researchers worldwide to extend mechanistic interpretability methods to safety-critical domains. The work builds directly on the need for concrete circuit-level analyses highlighted in "A Practical Review of Mechanistic Interpretability for Transformer-Based Language Models" and the broader landscape of explainability described in "Explainability of Large Language Models: Opportunities and ...". In addition, we will organize a yearly workshop at the university's AI safety symposium, inviting scholars from academia and industry to share best practices for safe model deployment. These activities will accelerate the development of transparent AI systems and provide a reusable foundation for future safety-oriented interpretability research. (2) LOCAL impact: At our institution, the project will support the undergraduate research program in Computer Science by providing hands-on training for students in SAE training, activation patching with TransformerLens, and causal analysis of model behavior. Approximately a dozen undergraduates each year will work on the open-source pipeline as part of a semester-long research experience, with a focus on first-generation and underrepresented minority students recruited through the university's Diversity in Computing Initiative. We will embed the project's methods into the graduate-level course "Mechanistic Interpretability of Large Language Models", giving students practical experience with state-of-the-art tools. Moreover, we will mentor a small cohort of high-school interns from local STEM outreach programs, introducing them to the concepts of AI safety and interpretability through workshops that culminate in a showcase of their findings. These concrete educational activities align with NSF's expectation that broader impacts articulate who benefits and how, and they follow the structured approach recommended by the MIT Communication Lab CommKit for effective broader-impacts narratives.

3 Introduction and Motivation

Recent incidents of language models producing plausible but false statements have highlighted a concrete safety failure mode: strategic deception. When a model deliberately frames a misleading answer as truthful, users of high-stakes applications—such as medical-advice chatbots, financial-service assistants, or policy-analysis tools—can be led to make harmful decisions. The TruthfulQA benchmark, which aggregates 817 questions spanning 38 categories of known human falsehoods, demonstrates that even state-of-the-art models frequently generate deceptive completions on topics where accuracy is critical. Mechanistic interpretability seeks to expose the internal circuitry that gives rise to such behavior. As noted in "A Practical Review of Mechanistic Interpretability for Transformer-Based Language Models," the field distinguishes itself by probing

the model’s internals rather than relying on input-output saliency alone. Existing surveys (e.g., “From Understanding to Utilization: A Survey on Explainability for Large Language Models” and “Explainability of Large Language Models: Opportunities and ...”) describe two dominant strands—circuit discovery and causal tracing—but they focus on generic feature analysis or post-hoc labeling. None of these works provide a method for identifying the minimal set of circuits that become active specifically during deceptive or strategically misleading responses, nor do they offer a causal ablation mechanism that can be applied at inference time. Because the current toolkit stops at interpretation without intervention, developers cannot reliably suppress deceptive behavior, and alignment techniques such as RLHF have shown limited success in eliminating strategically misleading outputs. Closing this gap would give the AI safety community a direct, mechanistic handle on deception: by pinpointing the sparse autoencoder (SAE) features that correlate with deceptive token patterns and ablating the associated feed-forward neurons, we can reduce the frequency of harmful completions while preserving overall model performance. This capability is essential for deploying trustworthy language-model systems in domains where misinformation carries real-world costs.

4 Novelty and Relation to Prior Work

Recent mechanistic interpretability research has established two complementary strands. First, circuit discovery approaches have shown that individual neurons and small groups of feed-forward layers can be interpreted as functional modules (see A Practical Review of Mechanistic Interpretability for Transformer-Based Language Models and From Understanding to Utilization: A Survey on Explainability for Large Language Models). Second, feature-level analyses have demonstrated that sparse autoencoders (SAEs) can extract highly interpretable latent features from the residual-stream activations of large language models (Explainability of Large Language Models: Opportunities and ...). Together, these works provide a powerful toolbox for probing model internals, but they stop short of a causal, circuit-level intervention that targets a concrete safety failure mode. The specific gap we aim to fill is three-fold. (1) Existing SAE studies focus on post-hoc labeling of features; they do not identify the minimal set of features that are causally responsible for deceptive or strategically misleading outputs. (2) Prior circuit-discovery papers have not leveraged SAEs to locate such features, nor have they demonstrated a systematic ablation protocol that preserves overall model competence. (3) No work has combined SAE-derived feature decomposition with targeted ablation to mitigate deceptive behavior at inference time, especially on a benchmark that explicitly measures truthfulness (TruthfulQA). Our proposal will therefore be the first to (i) use SAE-derived representations to pinpoint a compact circuit that is active during deceptive completions, (ii) apply causal ablation (via activation patching) to this circuit, and (iii) evaluate the resulting trade-off between deception reduction and language modeling quality. By closing this gap, we will provide a concrete mechanistic handle on a safety-critical failure mode that is not addressable by current alignment techniques such as RLHF. In short, while the literature has established the feasibility of both circuit discovery and SAE-based feature analysis, it has not yet demonstrated a minimal-circuit identification and causal mitigation strategy for deceptive behavior. Our work directly addresses this unfilled niche.

5 Project Goal

The goal of this project is to address the following research gap: Sparse autoencoders (SAEs) can decompose LLM residual-stream activations into interpretable features, but existing work focuses on post-hoc labeling—no method currently identifies the minimal circuits active specifically during deceptive or strategically misleading responses, nor provides a causal ablation mechanism to mitigate such behavior at inference time. Specifically, we test whether Applying SAE-derived min-

imal-circuit identification and targeted ablation will reduce the frequency of deceptive completions on a curated benchmark of strategically misleading prompts by $\geq 30\%$ relative to the unmodified model, while increasing overall perplexity by $< 5\%$. (H1). We further examine the underlying mechanism: The observed reduction is caused by the removal of a small set ($\leq 5\%$ of total SAE features) that show high activation overlap with token-level “deception” patterns; these features map onto specific feed-forward neurons whose ablation directly lowers the logits for deceptive tokens, as demonstrated by a causal intervention analysis (e.g., do-calculus or counterfactual ablation). (H2).

Testable Hypotheses

- H1 (Outcome): Applying SAE-derived minimal-circuit identification and targeted ablation will reduce the frequency of deceptive completions on a curated benchmark of strategically misleading prompts by $\geq 30\%$ relative to the unmodified model, while increasing overall perplexity by $< 5\%$.
- H2 (Mechanism): The observed reduction is caused by the removal of a small set ($\leq 5\%$ of total SAE features) that show high activation overlap with token-level “deception” patterns; these features map onto specific feed-forward neurons whose ablation directly lowers the logits for deceptive tokens, as demonstrated by a causal intervention analysis (e.g., do-calculus or counterfactual ablation).
- H3 (Impact): The identified minimal circuits and the ablation protocol will generalize across model scales (e.g., 1.3B vs. 6B parameters) and to out-of-distribution deceptive prompts, yielding a Pearson correlation $r \geq 0.6$ between the count of activated deception-associated SAE features and the measured deception rate across ≥ 3 distinct prompt families.

6 Methods: Technical Approach and Agent Workflow

Method Overview – We will combine sparse autoencoder (SAE) feature discovery with causal activation patching to (i) locate minimal circuits that are systematically active during deceptive completions, and (ii) intervene on those circuits to reduce deceptive output while preserving overall language modeling quality. The workflow is designed to be fully reproducible: all data, code, and random seeds will be released. Step 1: Data preparation 1. Use the TruthfulQA benchmark (817 questions across 38 categories of known human falsehoods). 500 questions will serve as the ablation-training set; the remaining 317 questions form a held-out generalization set. 2. For each prompt we record the model’s residual-stream activations at every layer using TransformerLens. 3. We label each generated answer as deceptive or non-deceptive by applying the TruthfulQA rubric (answers that contain a known falsehood are marked deceptive). This labeling provides the target variable for downstream correlation analysis. Step 2: SAE training 1. Train a sparse autoencoder on the residual-stream activations collected in Step 1 using SAELens. The SAE architecture follows the configuration of Sparse Autoencoders Find Highly Interpretable Features in Language Models, which demonstrated that a bottleneck of 2% of the hidden dimension yields highly interpretable features. 2. The SAE is trained on the 500-question ablation-training set for 50k steps with a sparsity coefficient of 0.1, producing a dictionary of features (each a linear combination of activation dimensions). 3. Output of this step: a set of SAE features and their activation patterns for every token in every prompt. Step 3: Identifying deception-associated features 1. Compute the activation overlap between each SAE feature and the token-level deception labels (e.g., using pointwise mutual information). 2. Rank features by their PMI score and select the smallest subset that collectively accounts for at least 80% of the total PMI mass. 3. Map each selected feature back to the underlying feed-forward neuron(s) that contribute the most weight in the SAE decoder. 4. This yields a minimal circuit consisting of $\leq 5\%$ of the total SAE features, as required by Hypothesis H2.

Step 4: Causal ablation via activation patching 1. Using TransformerLens, we perform activation patching: for a given input we replace the activation of the identified feed-forward neurons with their mean activation (computed over the non-deceptive subset). 2. We evaluate the patched model on the held-out 317-question set, measuring (a) the frequency of deceptive completions and (b) the overall perplexity. 3. Success criteria: - Deceptive completion rate reduced by $\geq 30\%$ relative to the unmodified model (H1). - Increase in perplexity $< 5\%$ (H1). 4. We repeat the ablation experiment on a second architecture (Pythia-6.9B) to test cross-scale generality (H3). Step 5: Mechanistic validation 1. For each ablated feature we conduct a counterfactual intervention: we set the feature to zero and recompute the logits for the next token. 2. We quantify the change in logits for tokens that belong to the deceptive token set (as defined by the TruthfulQA rubric). 3. A statistically significant reduction ($p < 0.05$, paired t-test) across the held-out set will support the causal mechanism claim of H2. Step 6: Generalization analysis 1. We construct three additional prompt families (political misinformation, medical myth, and financial fraud) that are not present in TruthfulQA. 2. For each family we compute the count of activated deception-associated SAE features and the observed deception rate. 3. We calculate Pearson’s r ; a correlation ≥ 0.6 will satisfy H3. Reproducibility and Resources

- Code will be built on the open-source TransformerLens and SAELens libraries. - All hyperparameters (learning rates, sparsity coefficients, random seeds) are recorded in a configuration file. - Computations will be run on a single NVIDIA A100 GPU (24GB memory) for each model scale, which fits within the allocated compute budget. Relation to Prior Work
- Sparse Autoencoders Find Highly Interpretable Features in Language Models showed that SAEs can isolate semantically meaningful directions but did not address circuit-level identification of deceptive behavior. - A Practical Review of Mechanistic Interpretability for Transformer-Based Language Models frames circuit discovery and causal tracing as complementary approaches; our method concretely combines them to fill the gap identified in the proposal. By following this workflow, we directly address the gap that existing SAE work stops at post-hoc labeling and does not provide a causal mechanism for mitigating deceptive outputs.

7 System Overview

[Figure: System architecture or workflow diagram for mechanistic interpretability of language models.
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Figure 1: Proposed workflow architecture for mechanistic interpretability of language models. See Section 6 for detailed stage descriptions.

8 Expected Results and Research Milestones

This project targets the following milestones and expected outcomes:

- Milestone 1 (Weeks 1–3): Literature review, dataset acquisition, and baseline implementation.
- Milestone 2 (Weeks 4–6): Data collection and preprocessing using TruthfulQA (817 questions across 38 categories of known human falsehoods) (n=Full 817-question benchmark; 500 for

ablation training + 317 held-out for generalization testing).

- Milestone 3 (Weeks 7–9): Core implementation and initial evaluation against baselines.
- Milestone 4 (Weeks 10–12): Full evaluation of hypotheses, statistical analysis, and ablation studies.
- Milestone 5 (Weeks 13–14): Final write-up, figure polishing, and submission preparation.

9 Risks and Mitigation

Risks and Mitigation Plan

Technical feasibility – Identifying minimal circuits that drive deceptive completions with sparse autoencoders (SAEs) can be computationally intensive, especially when scaling from GPT-2 XL (1.5 B) to Pythia-6.9 B. We will mitigate this risk by first developing the pipeline on the smaller model, where SAELens and TransformerLens provide efficient activation-patching utilities. The code will be profiled and parallelized before moving to the larger model, and we will reserve dedicated GPU slots on the institution’s HPC cluster to avoid queue delays.

Data limitation – TruthfulQA, while a standard benchmark for factuality, may not capture the full spectrum of strategically misleading prompts. To reduce the risk of over-fitting to a narrow set of deception patterns, we will augment the benchmark with a hand-crafted set of deceptive prompts covering at least three distinct prompt families (e.g., “misinformation framing”, “ambiguous question”, “self-contradiction”). The 317-question hold-out set will be used to assess generalization, and any failure to transfer will trigger an iterative redesign of the prompt collection.

Over-ablation risk – Targeted ablation of SAE-derived features could degrade overall language-model performance. Our hypothesis H1 explicitly bounds this effect (perplexity increase $< 5\%$). We will therefore monitor perplexity on a separate validation split after each ablation step; if the increase exceeds the bound, the ablation will be rolled back and the feature selection criteria refined. This safeguard directly addresses the risk of harming unrelated capabilities.

Generalization across scales – Circuits identified on GPT-2 XL may not be present or may behave differently in larger models. To mitigate this, cross-scale validation is built into the project milestones (see timeline below). We will quantify the relationship between the count of deception-associated SAE features and the measured deception rate; if the Pearson correlation falls below the target ($r \geq 0.6$), we will explore alternative feature-selection thresholds or incorporate additional layers of the residual stream.

Project schedule risk – Unforeseen delays in compute allocation or data preparation could push milestones. The project is organized into three phases with built-in buffer weeks:

- Phase 1 (Months 1-3) – SAE training on GPT-2 XL, initial circuit discovery, and validation of the feature-selection pipeline.

- Phase 2 (Months 4-6) – Targeted ablation experiments on the TruthfulQA benchmark, measurement of deception reduction and perplexity impact.
 - Phase 3 (Months 7-9) – Cross-scale testing on Pythia-6.9 B, correlation analysis, and preparation of the final report. Weekly progress meetings and a mid-project review at the end of Phase 2 will allow early detection of slippage and re-allocation of resources.
- Resource risk** – Access to sufficient GPU hours is essential. We have secured a baseline allocation on the university’s GPU cluster and will supplement it with cloud credits if needed. This dual-source strategy ensures continuity of experiments. By explicitly enumerating these risks, providing concrete mitigation strategies, and aligning the timeline with the NSF and CS222 expectations (,), we demonstrate a realistic and well-resourced plan. The adequacy of our computational and software resources is also addressed in accordance with the NSF merit-review element on adequate resources ().

10 Resources, Tools, and Budget

The following resources are required for this project:

- Compute: Standard workstation with GPU access (university cluster or cloud credits) for model training and evaluation.
- Software: Python scientific stack (PyTorch/TensorFlow, scikit-learn, pandas), LaTeX for document preparation, Git for version control.
- Data: TruthfulQA (817 questions across 38 categories of known human falsehoods) (target n=Full 817-question benchmark; 500 for ablation training + 317 held-out for generalization testing). Instruments: TransformerLens (activation patching), SAELens (SAE training), GPT-2 XL 1.5B and Pythia-6.9B for cross-scale testing.
- Budget: No direct funding required; project uses free-tier API access and university compute resources.
- Release plan: Code and evaluation scripts will be released as an open-source repository upon project completion.

References

- [1] Unknown. A Practical Review of Mechanistic Interpretability for Transformer-Based Language Models (). <https://arxiv.org/abs/2407.02646>
- [2] Unknown. [PDF] Exploring Look-Ahead Planning Mechanistic Interpretability in Large ... (). <https://arxiv.org/abs/https://aclanthology.org/2024.emnlp-main.440.pdf>
- [3] Unknown. Mechanistic interpretability of large language models with applications to the financial services industry (). <https://arxiv.org/abs/https://arxiv.org/html/2407.11215v1>
- [4] Unknown. From Understanding to Utilization: A Survey on Explainability for Large Language Models (). <https://arxiv.org/abs/https://arxiv.org/html/2401.12874v2>
- [5] Unknown. Explainability of Large Language Models: Opportunities and ... (). <https://arxiv.org/abs/https://arxiv.org/html/2510.17256v1>
- [6] Unknown. A Practical Review of Mechanistic Interpretability for Transformer-Based Language Models (). <https://arxiv.org/abs/https://arxiv.org/html/2407.02646v4>